

Consider the following polyhedron represented in canonical form:

$$\mathcal{P} = \{x \in \mathbb{R}^2 \mid Ax \leq b\},$$

with

$$A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 4 \\ 0 \\ 0 \end{pmatrix}.$$

1. Write explicitly the inequality constraints.
2. Draw the polyhedron. Hint: each constraint defines a half-plane, and the intersection of the half-planes gives the polyhedron. Draw first the lines defined by the equality version of the constraints, and identify on which side each half-plane lies.
3. Write the polyhedron in standard form.